CMOR 451/551 Homework 1
Fall 2026

This homework must be completed without the use of generative AI tools. You may use your class notes, textbook, and standard programming tools. The purpose of this assignment is for you to work through the derivations, algorithms, and simulation logic yourself.

Please submit all code files used to complete the homework along with your written solutions.

1) (25 pts) Inverse transform method can be used to generate truncated distributions. Suppose that a continuous random variable X has the density f with corresponding distribution function F. Assume that F is strictly increasing. For $a < b$, we define the truncated density

$$f^*(x) = \begin{cases} \frac{f(x)}{F(b)-F(a)} & \text{if } a \le x \le b, \\ 0 & \text{otherwise} \end{cases}$$

which has corresponding truncated distribution function

$$F^*(x) = \begin{cases} 0 & \text{if } x < a, \\ \frac{F(x)-F(a)}{F(b)-F(a)}, & \text{if } a \le x \le b \\ 1 & \text{if } x > b. \end{cases}$$

Then the algorithm for generating an X having distribution function F^* is as follows:

1. Generate $U \sim U(0,1)$.
2. Let $V = F(a) + [F(b) - F(a)]U$.
3. Return $X = F^{-1}(V)$.

a) (11 pts) Show that the X defined by this algorithm has distribution function F^*.

b) (11 pts) Show that the following algorithm is also valid for generating X with distribution function F^* (assume again F is continuous and strictly increasing):

1. Generate $U \sim U(0,1)$.
2. If $F(a) \le U \le F(b)$, return $X = F^{-1}(U)$. Otherwise, go to step 1.

c) (3 pts) Which algorithm do you think is better in terms of computational effort (i.e., number of additions, multiplications, comparisons, etc.)?

2) (20 pts) Let X be an exponential random variable with mean 1.

a) (8 pts) Give an efficient algorithm for simulating a random variable whose distribution is the conditional distribution of X given that $X < 0.05$.

b) (8 pts) Generate 1000 such variables and use them to estimate $E[X|X < 0.05]$.

c) (4 pts) Determine the exact value of $E[X|X < 0.05]$ and compare it with your answer in part b).

3) (25 pts) Let X have a bounded p.d.f. f over the finite domain [a,b] and define $\lambda = \max_X[f(x)]$. Let U and Y be independent and from $U(0,1)$ and $U(a,b)$, respectively, and let S denote the event $U \leq f(Y)/\lambda$.

a) (8 pts) Determine the success probability Pr(S).

b) (9 pts) Show that if S holds, then Y has the p.d.f. f.

c) (8 pts) Suppose one randomly generates X from the p.d.f.

$$f(x) = \begin{cases} (e^{-x+1} - 1)/(e-2), & 0 \leq x \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

using the described acceptance-rejection scheme. Determine the corresponding Pr(S).

4) (30 pts) Give inverse-transform, composition, and acceptance-rejection algorithms for generating from each of the following densities. Discuss which algorithm is preferable for each density. (Hint. First plot densities.)

a) (15 pts)

$$f(x) = \begin{cases} 3x^2/2, & \text{if } -1 \leq x \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

b) (15 pts) For $0 < a < 1/2$,

$$f(x) = \begin{cases} 0, & \text{if } x \leq 0, \\ \frac{x}{a(1-a)}, & \text{if } 0 \leq x \leq a \\ \frac{1}{1-a}, & \text{if } a \leq x \leq 1-a, \\ \frac{1-x}{a(1-a)}, & \text{if } 1-a \leq x \leq 1, \\ 0, & \text{if } 1 \leq x \end{cases}$$